

SILICON CHIPS

A transistor is a device that changes electronic signals. It is made from silicon, which is “doped” to give it different properties. For example, when electrons are removed, “holes” are left for electrons to flow into. Putting together layers of silicon, each with either an excess (n-type) or deficit (p-type) of electrons, creates transistors that can amplify or switch a current. Many transistors are integrated onto a single computer chip.

Basic transistor

In an n-p-n transistor, a layer with a deficit of electrons (p-type) is sandwiched between layers with excess electrons (n-type). Excess electrons can flow into the p-type region.

